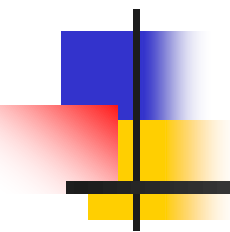




Indian Institute of Information Technology, Surat

भारतीय सूचना प्रौद्योगिकी संस्थान, सूरत
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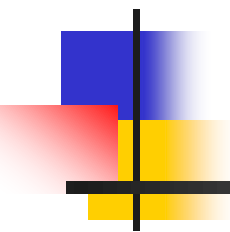
EC 503: Image Processing and Computer Vision

By:

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ECE Department

IIIT, Surat

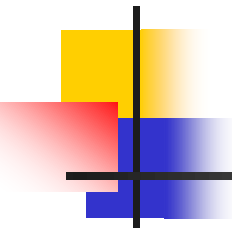


UNIT 2

Image Enhancement in Spatial Domain and Frequency Domain

UNIT2_1:

Image Enhancement in the Spatial Domain



Derivative operator

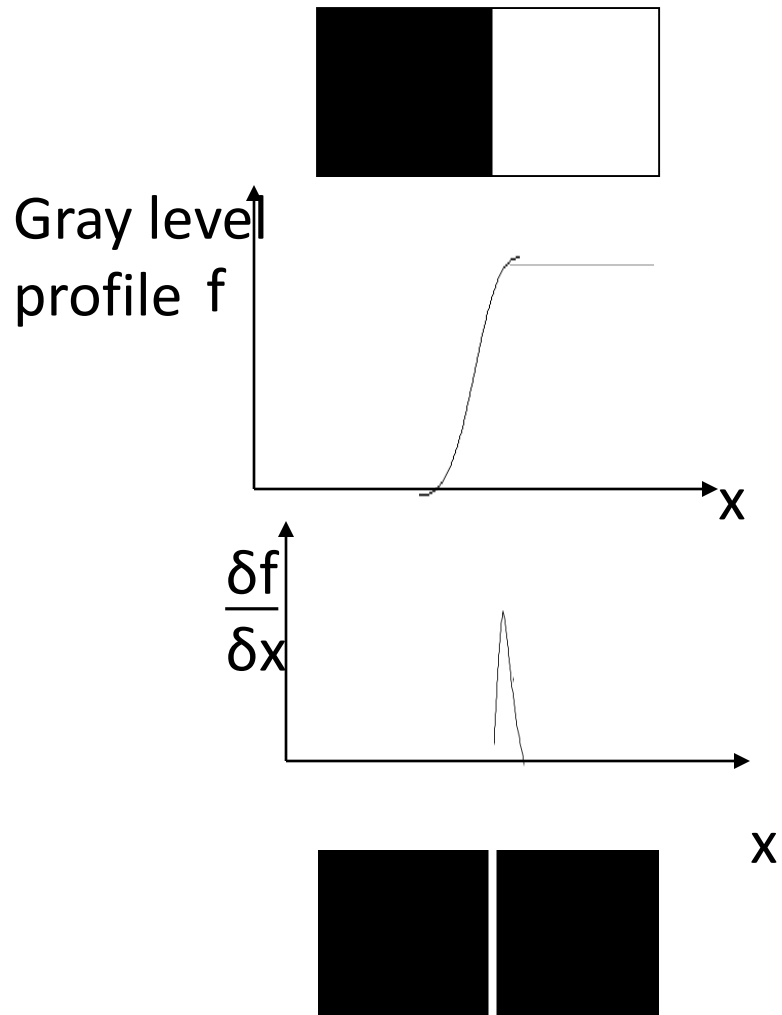
- The strength of the response of a derivative operator is proportional to the degree of operator is proportional to the degree of discontinuity of the image at the point at which the operator is applied
- Image differentiation
 - Enhances edges and other discontinuities (noise)
 - Deemphasizes area with slowly varying gray-level values



Sharpening – Spatial Differentiation

- Image differentiation enhances edges and noise and deemphasizes areas with slowly varying gray level values
- **Foundation:**
- 1st and 2nd order derivatives – defined in terms of differences
- **Definition for first derivative**
- Must be zero – in flat segments
- Must be non zero – at on set of a gray level step or ramp
- Must be non zero – along ramp
- **Definition for second derivative**
- Must be zero – in flat segments
- Must be non zero – at on set and end of gray level step or ramp
- The shortest distance over which change can occur is between adjacent pixels

Fundamental of derivatives in image sharpening

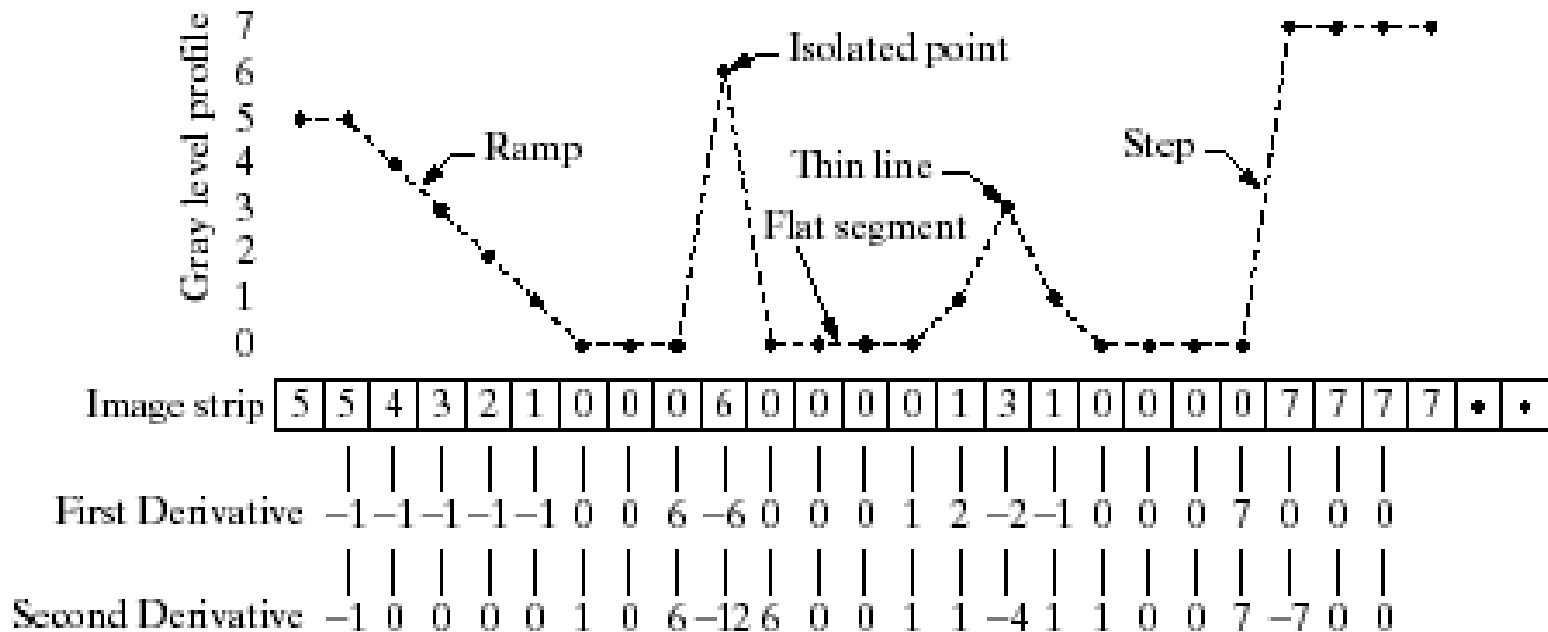


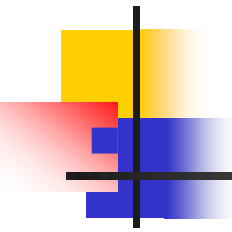
Sharpening Spatial Filters

■ Foundation:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

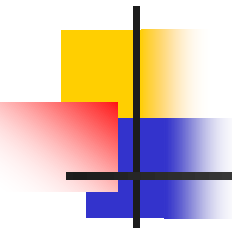
$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$





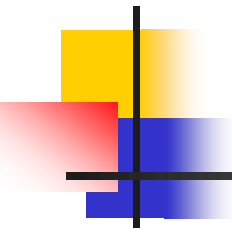
Observation

- 1st order derivative is nonzero along the entire ramp, while 2nd order derivative is nonzero only at the onset and end of ramp.
- 1st order derivative produce thick edges and 2nd order derivative much finer ones.
- Response at isolated noise point is much stronger for 2nd order derivative hence it can enhance fine details much more than 1st order derivative.
- For step change 1st order derivative generally have a stronger response.
- 2nd order derivative produces a double response to a gray level step (-ve to +ve)
- The second derivative is better suited than the first derivative for image enhancement because of the ability of the former to enhance fine detail.



Isotropic filter

- The filter having the response is independent of the direction of the discontinuities in the image to which the filter is applied
- In other words isotropic filters are **rotation invariant**, in the sense that rotating the image and then applying the filter gives the same result as applying the filter first and then rotate the image



First and Second-order derivative of 2D

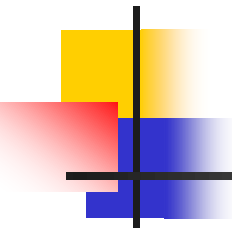
- When we consider an image function of two variables, $f(x, y)$, at which time we will be dealing with partial derivatives along the two spatial axes

Gradient operator
(linear operator)

$$\nabla f = \frac{\partial f(x, y)}{\partial x \partial y} = \frac{\partial f(x, y)}{\partial x} + \frac{\partial f(x, y)}{\partial y}$$

Laplacian operator
(non-linear)

$$\nabla^2 f = \frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2}$$

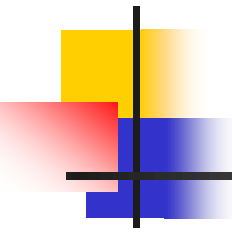


Discrete form of Laplacian

from $\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$

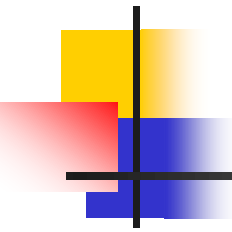
$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\begin{aligned} \nabla^2 f = & [f(x+1, y) + f(x-1, y) \\ & + f(x, y+1) + f(x, y-1) - 4f(x, y)] \end{aligned}$$



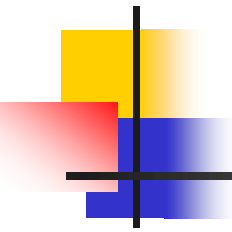
Result Laplacian mask

0	1	0
1	-4	1
0	1	0



Laplacian mask implemented an extension of diagonal neighbors

1	1	1
1	-8	1
1	1	1

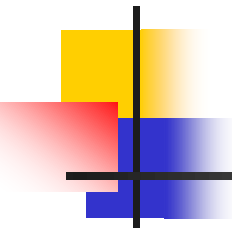


Other implementation of Laplacian masks

- Give the same result, but we have to keep in mind that when combining (add / subtract) a Laplacian-filtered image with another image

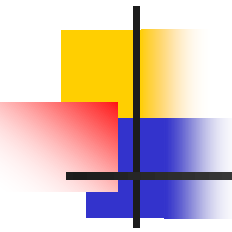
0	-1	0
-1	4	-1
0	-1	0

-1	-1	-1
-1	8	-1
-1	-1	-1



Effect of Laplacian Operator

- As it is a derivative operator
 - It highlights gray-level discontinuities in an image
 - It deemphasizes regions with slowly varying gray levels
- Tends to produce images that have
 - Grayish edge lines and other discontinuities, all superimposed on a dark
 - Featureless background



Correct the effect of featureless background

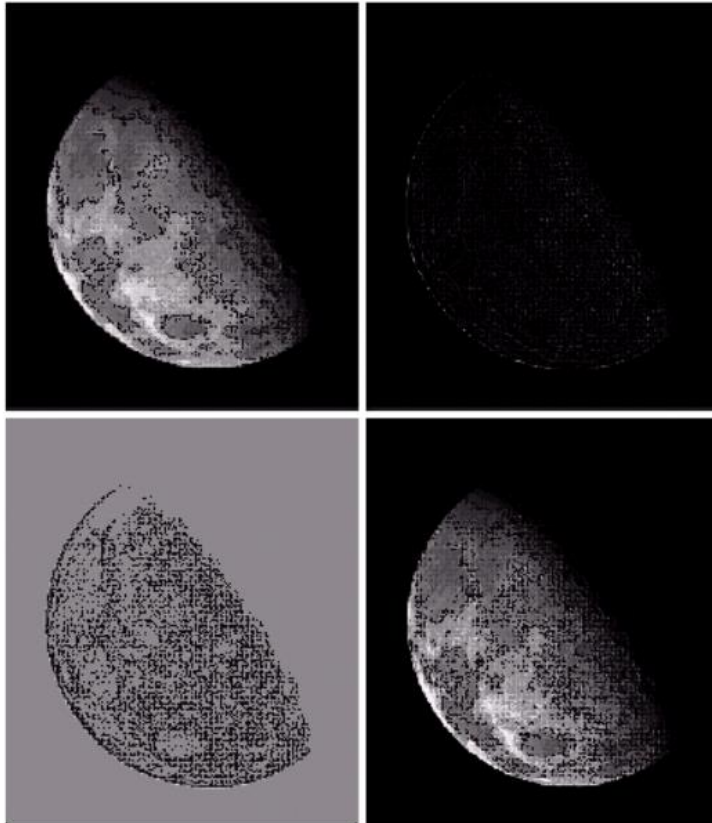
- Easily by adding the original and Laplacian image
- Be careful with the Laplacian filter used be careful with the Laplacian filter used

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) \\ f(x, y) + \nabla^2 f(x, y) \end{cases}$$

if the center coefficient
of the Laplacian mask is
negative

if the center coefficient
of the Laplacian mask is
positive

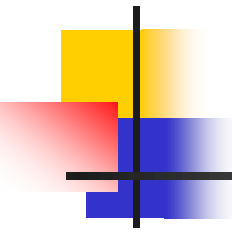
Example



- a). image of the North pole of the moon
- b). Laplacian-filtered image with

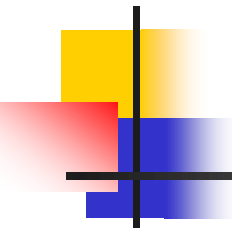
1	1	1
1	-8	1
1	1	1

- c). Laplacian image scaled for display purposes
- d). image enhanced by addition with original image



Mask of Laplacian + addition

- To simplify the computation, we can create a mask which do both operations, Laplacian Filter and Addition the original image



Mask of Laplacian + addition

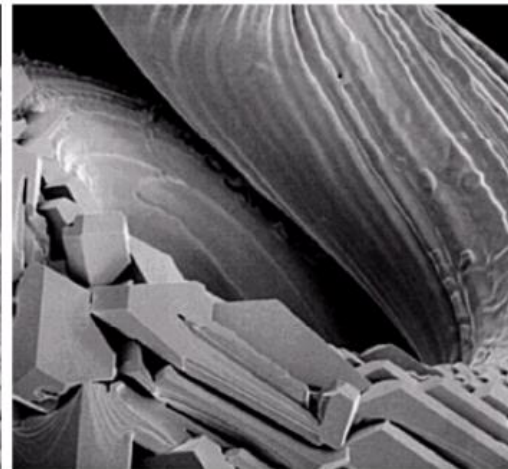
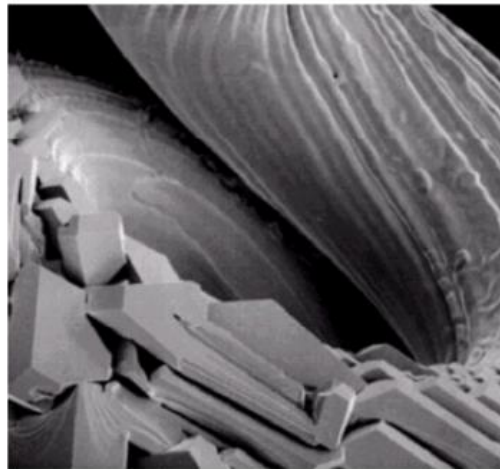
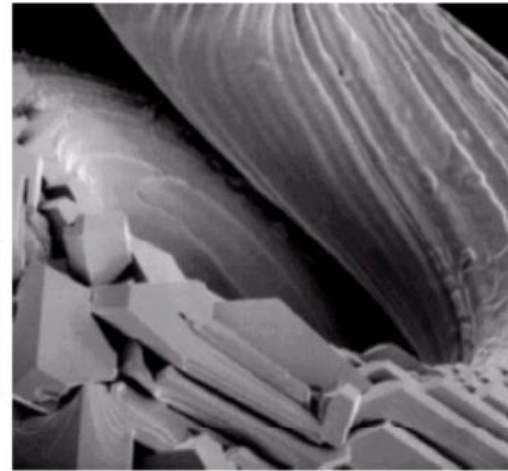
$$\begin{aligned} g(x, y) &= f(x, y) - [f(x+1, y) + f(x-1, y) \\ &\quad + f(x, y+1) + f(x, y-1) + 4f(x, y)] \\ &= 5f(x, y) - [f(x+1, y) + f(x-1, y) \\ &\quad + f(x, y+1) + f(x, y-1)] \end{aligned}$$

0	-1	0
-1	5	-1
0	-1	0

Example

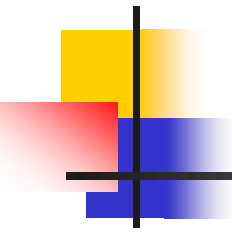
0	-1	0
-1	5	-1
0	-1	0

-1	-1	-1
-1	9	-1
-1	-1	-1



a b c
d e

FIGURE 3.41 (a) Composite Laplacian mask. (b) A second composite mask. (c) Scanning electron microscope image. (d) and (e) Results of filtering with the masks in (a) and (b), respectively. Note how much sharper (e) is than (d). (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)



Note

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) \\ f(x, y) + \nabla^2 f(x, y) \end{cases}$$

0	-1	0
-1	5	-1
0	-1	0

=

0	0	0
0	1	0
0	0	0

+

0	-1	0
-1	4	-1
0	-1	0

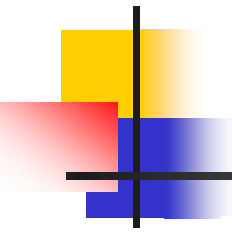
0	-1	0
-1	9	-1
0	-1	0

=

0	0	0
0	1	0
0	0	0

+

0	-1	0
-1	8	-1
0	-1	0

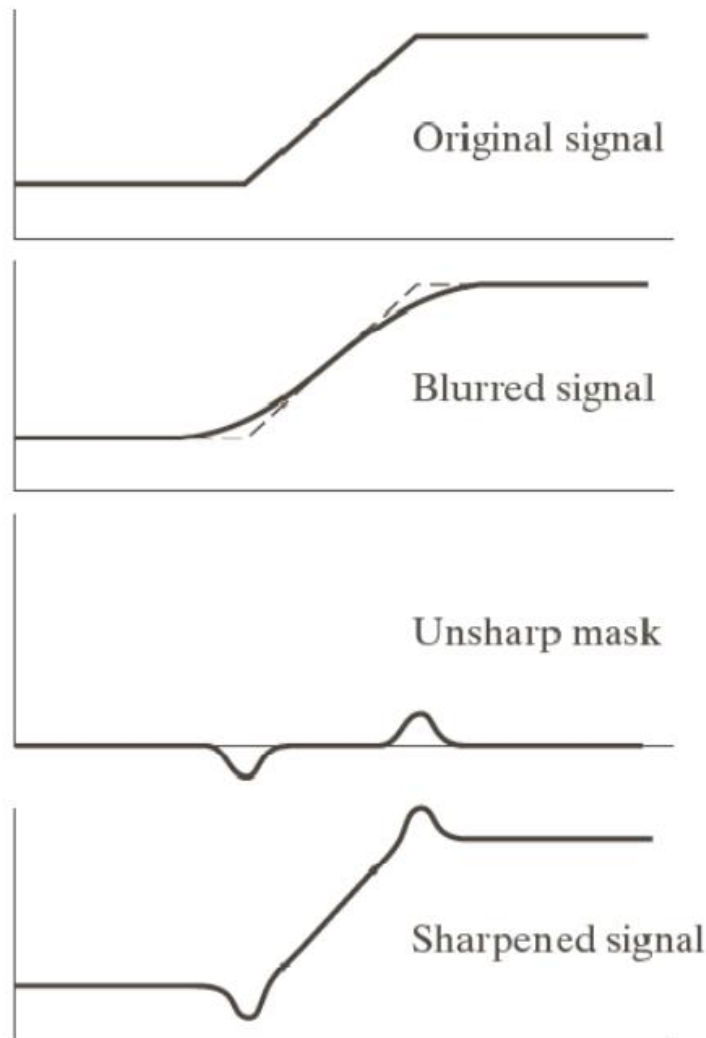


Unsharp masking

$$f_s(x, y) = f(x, y) - \bar{f}(x, y)$$

- Sharpened image = original image – blurred image
- To subtract a blurred version of an image produces sharpening output image

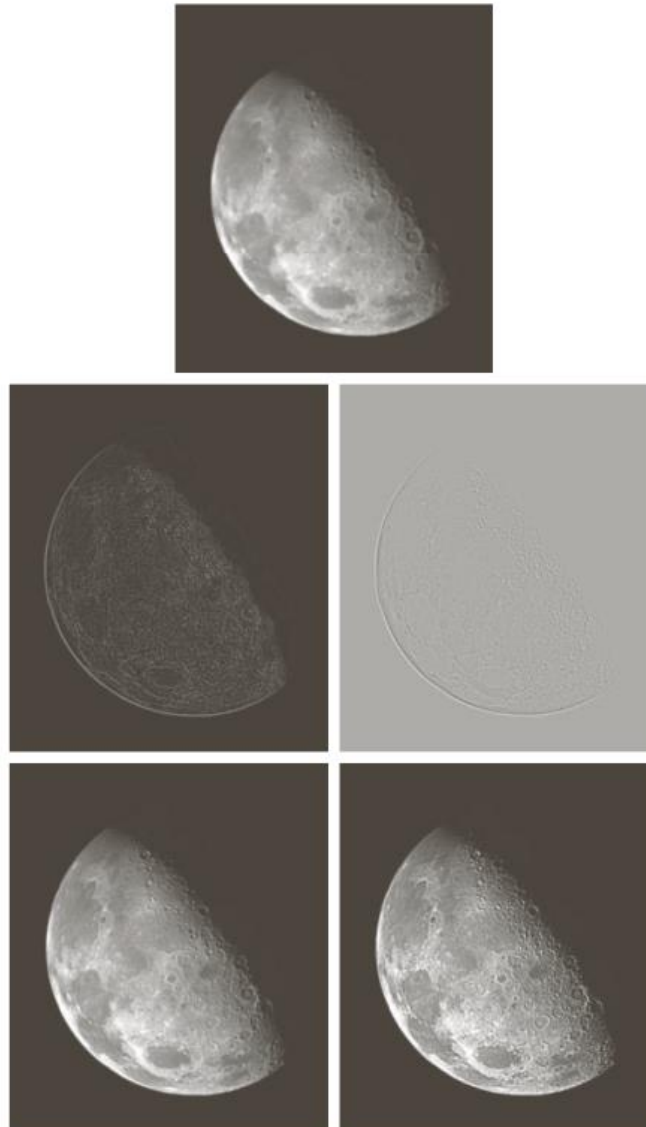
Unsharp mask



a
b
c
d

FIGURE 3.39 1-D illustration of the mechanics of unsharp masking. (a) Original signal. (b) Blurred signal with original shown dashed for reference. (c) Unsharp mask. (d) Sharpened signal, obtained by adding (c) to (a).

Example



a	
b	c
d	e

FIGURE 3.38

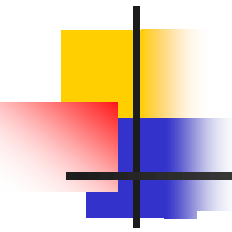
(a) Blurred image of the North Pole of the moon.

(b) Laplacian without scaling.

(c) Laplacian with scaling. (d) Image sharpened using the mask in Fig. 3.37(a).

(e) Result of using the mask in Fig. 3.37(b).

(Original image courtesy of NASA.)

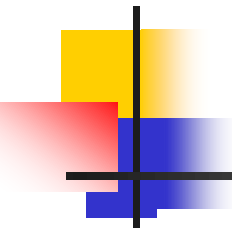


High-boost filtering

$$f_{hb}(x, y) = Af(x, y) - \bar{f}(x, y)$$

$$\begin{aligned} f_{hb}(x, y) &= (A-1)f(x, y) - f(x, y)\bar{f}(x, y) \\ &= (A-1)f(x, y) - f_s(x, y) \end{aligned}$$

- Generalized form of Unsharp masking
- $A \geq 1$



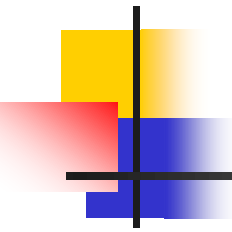
High-boost filtering

$$f_{hb}(x, y) = (A - 1)f(x, y) - f_s(x, y)$$

- If we use Laplacian filter to create sharpen image $f_s(x, y)$ with addition of original image

$$f_s(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) \\ f(x, y) + \nabla^2 f(x, y) \end{cases}$$

$$f_{hb}(x, y) = \begin{cases} Af(x, y) - \nabla^2 f(x, y) \\ Af(x, y) + \nabla^2 f(x, y) \end{cases}$$



High boost masks

0	-1	0	-1	-1	-1
-1	$A + 4$	-1	-1	$A + 8$	-1
0	-1	0	-1	-1	-1

- $A \geq 1$
- If $A = 1$, it becomes “standard” Laplacian sharpening

Example

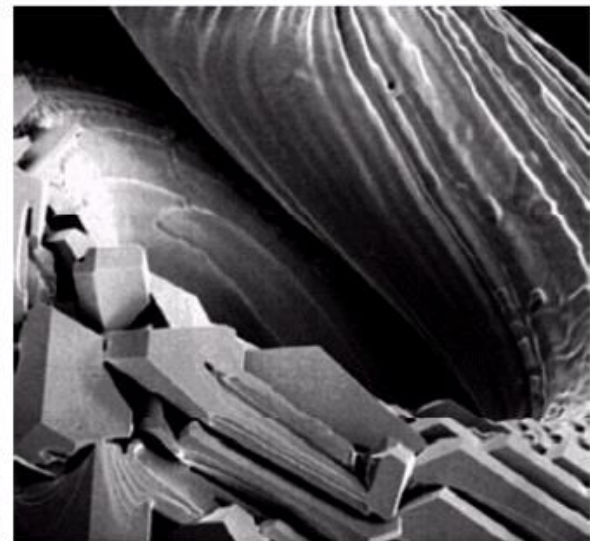
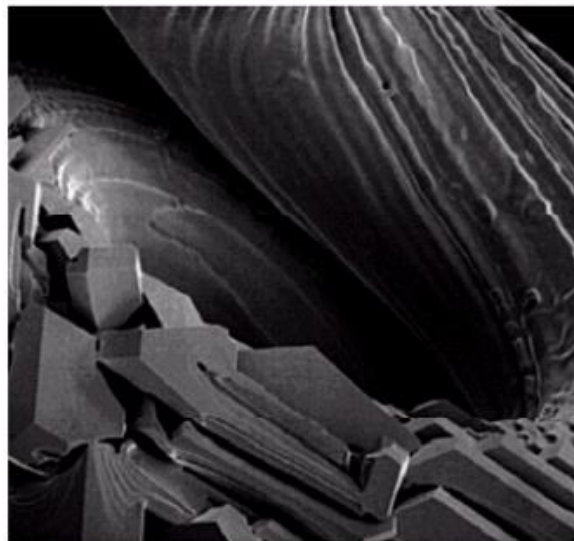
a	b
c	d

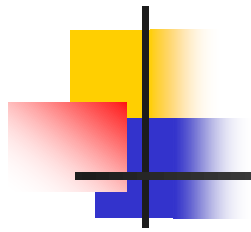
FIGURE 3.43

(a) Same as Fig. 3.41(c), but darker.

(b) Laplacian of (a) computed with the mask in Fig. 3.42(b) using $A = 0$.

(c) Laplacian enhanced image using the mask in Fig. 3.42(b) with $A = 1$. (d) Same as (c), but using $A = 1.7$.





THANK YOU